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Day Month Year

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Status of Thesis

Note: red = pre first edit, orange = pre second edit and beyond, green = done, blue = may not make it into the thesis, or it may be moved to the appendix.

- Chapter ??: Not started. Will write last/almost last
- Chapter ??: Some ideas for content, but no words written yet.
- Chapter ??: Approaching first draft. Lots of work is still required to make clear that terms are properly defined.
- Chapter 2: Vague idea for structure, but almost no words have been written.
- Chapter ??: Passed first drafting. Still to apply Ken and Bryan's edits (see email from 29/09/2020 'quick question' for notes).
- Chapter ??: Passed first drafting. Awaiting Ken and Bryan's comments.
- Chapter ??: Passed first drafting. Still to apply Ken and Bryan's edits.
- Chapter ??: Not written or properly planned
- Chapter ??: Not written or properly planned

- Chapter ?? : Not written or properly planned. Might not make it past planning stage
- Chapter ?? : Not written or properly planned. May be relegated to the appendix
- Chapter ?? : Significant amounts of thought has already gone into this.
- Chapter ?? : Not started. Will write last/almost last
- Appendix ?? : Passed first drafting.
- Appendix ?? : Mostly written.
- Appendix ?? : Mostly written, but needs writing in the notation of Chapter ??
- Appendix ?? : Mostly written
- Appendix ?? : Likely to be culled

Declaration

Acknowledgements

Chapter 1

Abstract

Chapter 2

High Power Laser Prototype for LIGO

2.1 Introduction

LIGO requires high input power, nominally 125 W, to decrease the shot noise which limits its sensitivity at high frequencies ($> 100\text{Hz}$). The ‘High Power Oscillator’ (HPO) was designed to provide this power. The HPO was a ring cavity containing laser crystals and would be seeded by a 35W laser. This design went through extensive testing and refinement at the AEI [1], and was installed as part of the LIGO ‘Pre-Stabilised-Laser’ (PSL). However, due to jitter noise coupling to the differential arm motion signal (DARM) from the water cooling of the the crystals within the HPO (as well as one version being destroyed by accident) [2], the HPO was bypassed, and LIGO operated using the 35 W laser.

The use of a single pass amplifier was investigated as a replacement of the HPO, and for third observation run a Neovan-4S deployed to step the PSL power up from 35 W to 70 W [3]. The next stage of development was a complete redesign of the PSL based around using two upgraded versions of the Neovan-4S, the Neovan-4SHP. This amplifier showed promising results [4, 5], so a prototype PSL was constructed and tested at LIGO Livingston. This design will be installed as the new PSL for the start of O4 [2].

2.2 Properties of the LIGO PSL

A laser has five properties: intensity, frequency, beam shape, pointing and polarisation. The PSL must provide a beam which satisfies the requirements for each of these properties [6]. Along with these basic properties of a laser beam, the PSL must also be able to be operated without interruption for timescales of the order 1 year, have remote diagnostic tools, and have a remote interface so that operators can control and monitor the status of the laser.

2.2.1 Intensity

The PSL should deliver 125 W to the interferometer. The intensity fluctuations should be lower than the noise generated by radiation pressure noise causing motion of the test masses. The intensity noise should be limited by shot noise around 100 Hz where radiation pressure noise has dropped off. Above the bandwidth of the cavities, the intensity noise requirement can be relaxed as the interferometer becomes less sensitive to intensity noise.

There are several ways of stabilising a laser. Modulation of one of the amplifiers pump diodes and intensity actuation by way of an acoustic optical modulator (AOM) were investigated during the prototyping of the new PSL design. By modulating the one of the pump light diodes for one of the amplifiers, we are able to reduce the amount of transmissive optics in the high power path of the beam by one. This is an attractive idea since the layouts are simpler to set-up and have less high power components. The AOM method is the one currently in use for intensity stabilisation at LIGO.

2.2.2 Frequency

Ultimately, LIGO uses the arm cavity as a frequency reference since $\delta f/f = \delta l/l$, however the PSL must provide a suitably low frequency noise to the interferometer so that it can acquire lock. The PSL uses an ultra low expansion fused silica reference cavity in a Pound-Drever-Hall (PDH) locking scheme to provide this frequency stabilisation [7].

2.2.3 Beam Shape

The PSL produces a high quality TM00 beam by using a cavity known as the ‘pre mode cleaner’ (PMC). This is the last stage of the PSL before the beam is handed off to the rest of the interferometer. The higher order mode content of the beam produced by earlier stages of the PSL is therefore important since higher order modes will be rejected by the PMC.

2.2.4 Pointing

The pointing noise witnessed at LHO is one of the reasons why single pass amplifiers were chosen as the replacement of the HPO. Pointing noise manifests as intensity noise since a change in beam direction can be modelled as the introduction of higher order modes. These new amplifiers are cooled in such a way that the jitter on the PSL beam is reduced.

2.2.5 Polarisation

Highly linear polarised light is essential to the operation of LIGO, since the optical coatings are all optimised for a specific polarisation.

2.2.6 Other Aspects of the PSL

There are many practical requirements of the PSL. It is useful to diagnose the PSL remotely, so a diagnostic breadboard (DBB) [8] capable of making pointing, intensity and frequency noise measurements, as well as mode scans, was integrated into the design of the PSL. Remote operation and monitoring is also required, so slow control is provided by a Beckhoff system.

2.3 Set-up and Layout

- diagram of layout
- description of layout
- use 2 amplifiers and a PMC to get high power gaussian beam

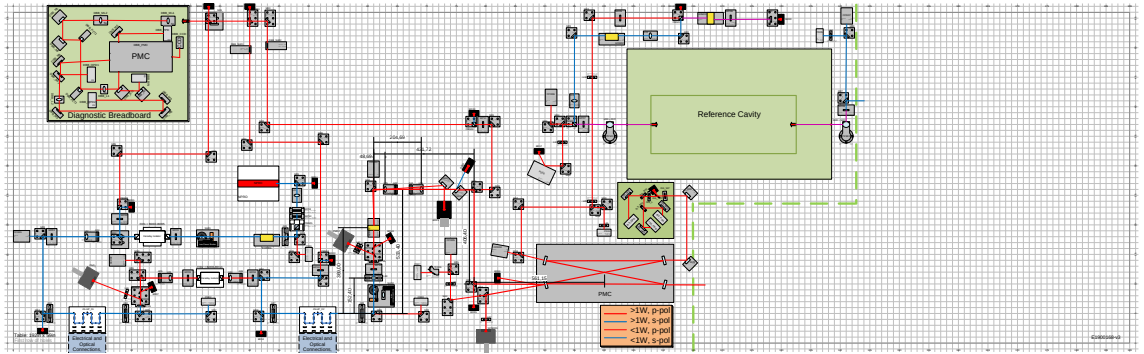


Figure 2.1: e1900168. Layout placeholder

- flow rates and beam quality of amplifiers

2.3.1 NPRO

The first element of the PSL is a non-planar ring oscillator laser [9, 10]. This laser consists of a single Nd:YAG crystal which acts as both the gain medium and the cavity, thus it produces a laser with high amounts of frequency stability.

2.3.2 OEM solid-state Amplifier

- why are they better than an HPO in principle
- how they work
- neovan 4S-HP

2.4 Frequency Noise

Need to put together the measurements for frequency noise.

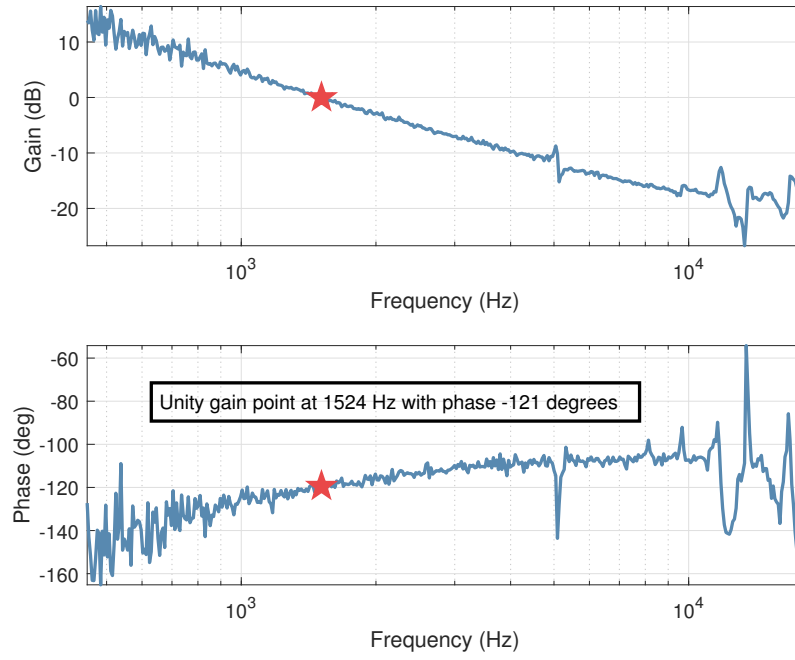


Figure 2.2: Open loop gain of frequency stabilisation loop

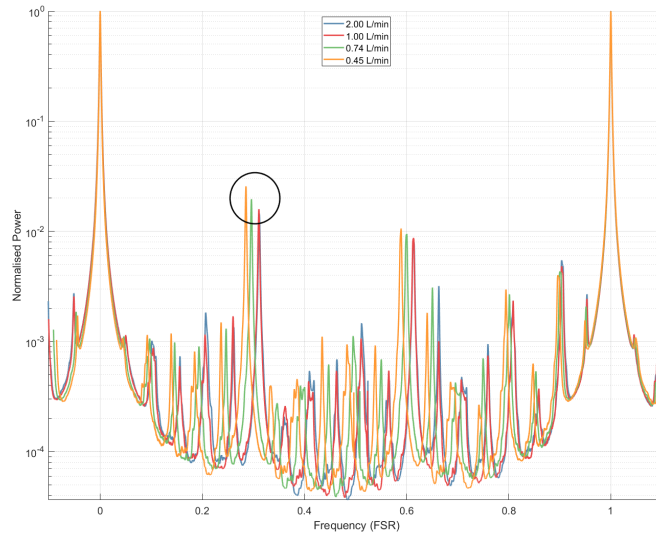


Figure 2.3: Flow rates affecting the thermal lensing, resulting in increased ellipticity

2.5 Conclusion

summary of how much power for how long, and how it could be improved slightly with mode matching

Chapter 3

Amplitude Stabilisation of High Power Laser Prototype for LIGO

3.1 Introduction + Requirements

- shot noise requirement
- radiation pressure noise setting noise requirement
- inner+outer loop design
- could include calculation for mismatches in the IFO? Point absorbers etc.
- Patrick Kwee's Thesis.
- Useful ligo reference T010075

3.1.1 Intensity Noise

SRC dither measurement.Measurement 12/11/2019. Amount of imbal-

ance $\sim 3\%$ Terra Intensity noise injection method. LIGO-T010075-v3 states 1% arm imbalance for nominal value. LIGO Document T050036-v4 also states 1%, this is probably a better reference.

3.1.2 Coupling of Intensity Noise

In the GW band, laser intensity noise couples via differential arm finesse (i.e. losses) $\rightarrow$ difference in arm powers $\rightarrow$ radiation pressure fluctuations which look like gw signal. Would be nice to have an equation.

Power fluctuations in LO light $\rightarrow$ can't be distinguished from GW signal. Will be cancelled with BHD.

PRC passively filters intensity noise of laser. Bandwidth $\approx 1\text{Hz}$.

3.1.3 Control band fluctuations 0.1-10Hz

Creates radiation pressure $\rightarrow$ GW signal coupling.

3.1.4 RF noise

RF signals used to control the IFO $\rightarrow$ Noise at these frequencies degrades SNR $\rightarrow$ indirect coupling of noise to GW channel. PMC used for this. IMC cannot be used as EOM before this. NPRO is dominant RF noise source.

3.1.5 Inner Loop

3.1.6 Outer Loop

3.2 Free Running Noise and Noise Hunting

- free running noise of the laser
- what dominates the laser noise? NPRO
- noise sources such as mounts
- noise from PMC offset not being tuned
- 4Hz noise from Beckhoff

3.3 AOM Stabilisation

- description of AOM
- AOM TF
- AOM loop TF
- plot of AOM stabilisation working

3.4 Current Shunt Stabilisation

- description of CS
- CS TF

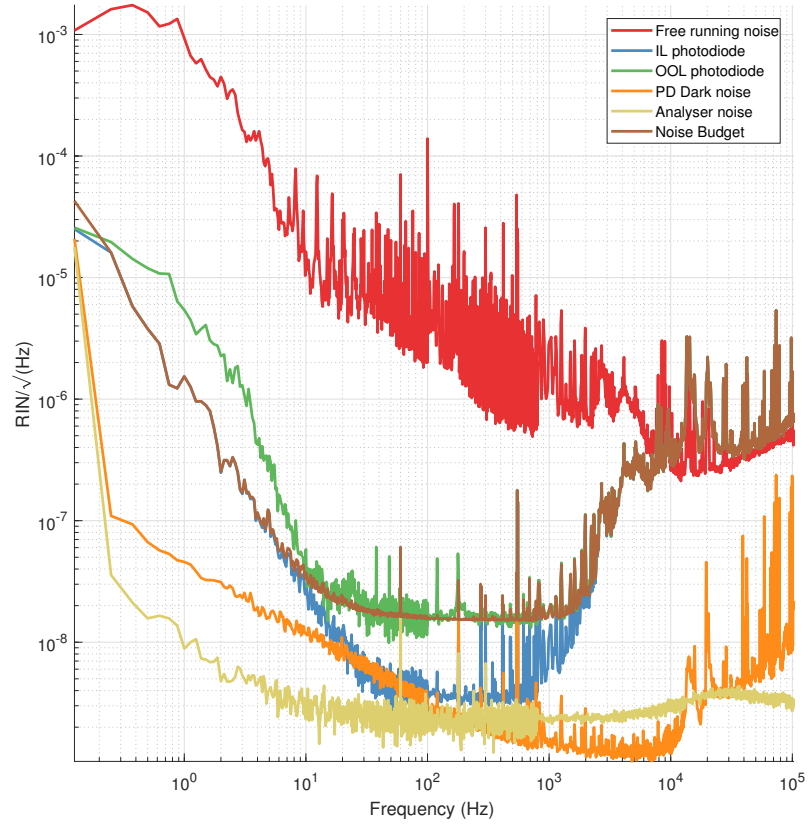


Figure 3.1: Free running and stabilised noise of laser

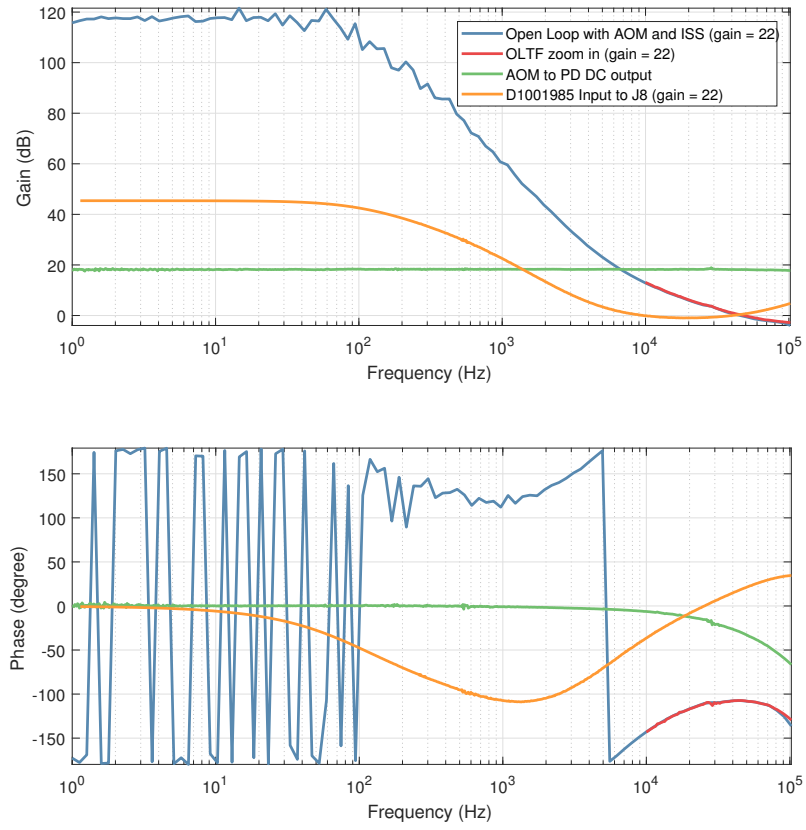


Figure 3.2: Transfer functions for parts of the AOM servo

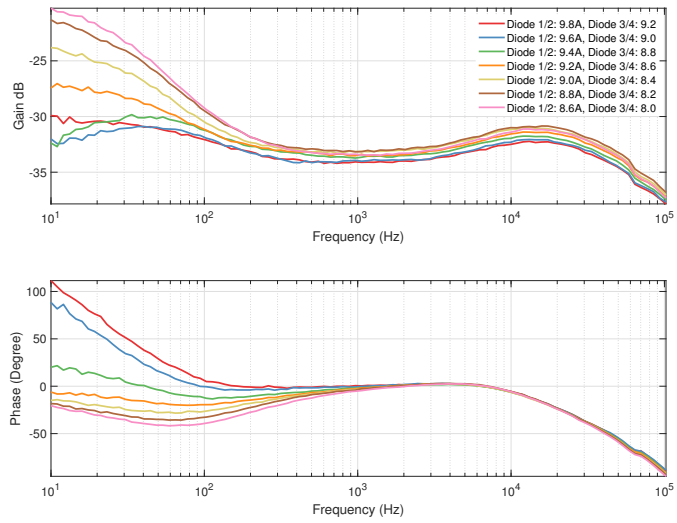


Figure 3.3: current shunt input to light power transfer function for different diode currents

- CS loop TF
- plot of CS stabilisation working

3.5 Comparison of Two stabilisation Schemes

- why I think the AOM is a better choice even though they both work

3.6 Conclusion

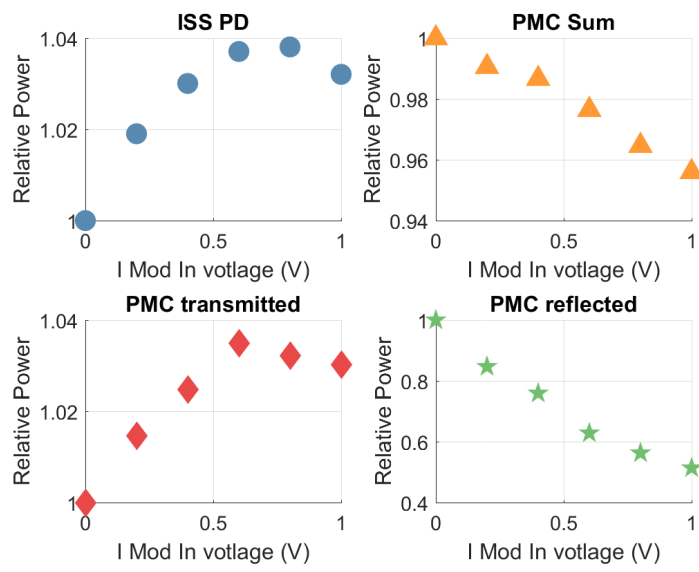


Figure 3.4: DC behaviour of current shunt

Bibliography

- [1] P. Kwee, C. Bogan, K. Danzmann, M. Frede, H. Kim, P. King, J. Pödl, O. Puncken, R. L. Savage, F. Seifert, P. Wessels, L. Winkelmann, and B. Willke. Stabilized high-power laser system for the gravitational wave detector advanced LIGO. *Optics Express*, 20(10):10617, apr 2012.
- [2] the LIGO Scientific Collaboration. Instrument science white paper 2019. LIGO internal document LIGO-T1900409–v3.
- [3] Andrew P. Spencer. *Advanced Techniques in Laser Interferometry for Current and Future Gravitational Wave Detectors*. PhD thesis, University of Glasgow, August 2019.
- [4] Fabian Thies, Nina Bode, Patrick Oppermann, Maik Frede, Bastian Schulz, and Benno Willke. Nd:YVO4 high-power master oscillator power amplifier laser system for second-generation gravitational wave detectors. *Optics Letters*, 44(3):719, jan 2019.
- [5] Nina Bode, Fabian Meylahn, and Benno Willke. Sequential high power laser amplifiers for gravitational wave detection. *Optics Express*, 28(20):29469, sep 2020.

- [6] B. Willke, P. King, R. Savage, and P. Fritschel. Pre-stabilized laser design requirements. DCC LIGO-T050036-v4, July 2011.
- [7] H. Kim, P. King, C. Krämer, P. Kwee, J. Pöld, R. Savage, P. Weßels, and B. Willke. Psl final design. DCC LIGO T0900649-v4, February 2010.
- [8] P. Kwee. *Laser Characterization and Stabilization for Precision Interferometry*. PhD thesis, Max Planck Institute for Gravitational Physics, 2010.
- [9] Thomas J. Kane and Robert L. Byer. Monolithic, unidirectional single-mode nd:YAG ring laser. *Optics Letters*, 10(2):65, feb 1985.
- [10] I Freitag, A Tünnermann, and H Welling. Power scaling of diode-pumped monolithic nd:YAG lasers to output powers of several watts. *Optics Communications*, 115(5-6):511–515, apr 1995.